

OC IT Projects — Chapter 9

Routing — Review Sheet + Full Quiz

Based on the Chapter 9 PowerPoint content.

1. Key Points

- Routing: the router connects two or more networks and determines the next network point to which data should be forwarded.
- Routing Table: contains entries for networks the router knows about and the path/interface toward the destination.
- Directly Connected Network: directly connected to the router.
- Remote Network: connected to another router.
- Routing Process: (1) De-encapsulate the Layer 3 packet by removing the Layer 2 frame header/trailer. (2) Examine the destination IP and find the best path. (3) Encapsulate into a new Layer 2 frame and forward through the exit interface.
- Static Routing: administrator manually constructs routing-table entries; static routes to network destinations are unchangeable.
- Dynamic Routing: routing tables are built and maintained automatically through communication between routers using routing protocols.
- Routing Metrics: measurable values used to differentiate routes. Examples: hop count, delay, bandwidth, load, reliability. The training states the best path is based on the lowest cost.
- Load Balancing: with two or more equal-cost paths, packets can use both paths.
- RIP: uses hop count. The training states that updates are broadcast to 255.255.255.255 every 30 seconds.
- Training classification: Distance-vector → RIP, EIGRP; Link-state → OSPF; Hybrid → BGP.

2. What You Got Wrong

- Q4: Directly Connected = directly attached to the router; Remote = reached through another router.
- Q23: For a large network that changes frequently, Dynamic Routing is the suitable concept because routers exchange routing information and maintain tables automatically.
- Core contrast: Static = manual configuration; Dynamic = automatic exchange using routing protocols.

3. Quick Memory Table

Concept	Remember
Directly Connected	Directly attached to router
Remote	Reached through another router
Static	Manual administrator configuration
Dynamic	Automatic routing-table exchange/maintenance
Metric	Used to compare routes
RIP	Hop count
Equal-cost Load Balancing	Uses multiple equal-cost paths
Distance-Vector	RIP, EIGRP
Link-State	OSPF
Hybrid (training)	BGP

4. Complete Quiz — All 30 Questions

Q1. What is the primary purpose of a router?

- A) Connect devices within the same LAN
- B) Connect two or more networks and determine where data should be forwarded
- C) Convert domain names to IP addresses
- D) Encrypt network traffic

Q2. What information does a router examine to determine where to forward an IP packet?

- A) Destination IP address
- B) Source MAC address only
- C) SSID
- D) Port number only

Q3. What is a routing table?

- A) A table containing MAC addresses only
- B) A list of wireless networks
- C) Information used by a router to determine paths to networks
- D) A list of open TCP ports

Q4. What is a directly connected network?

- A) A network reachable through another router
- B) A network directly attached to the router
- C) A network learned through RIP only
- D) A network available through the Internet

Q5. What is a remote network?

- A) A network directly connected to the router
- B) A network that exists behind another router
- C) A wireless-only network
- D) A network without a default gateway

Q6. When a router receives a Layer 2 frame, what happens first according to the training?

- A) It broadcasts the frame
- B) It changes the destination IP
- C) It de-encapsulates the Layer 3 packet
- D) It sends the packet to DNS

Q7. After examining the destination IP, what does the router do?

- A) Determines the best path from the routing table
- B) Changes the destination IP
- C) Creates a new IP address
- D) Converts IPv4 to IPv6

Q8. After choosing the exit interface, what does the router do?

- A) Deletes the packet
- B) Re-encapsulates the packet into a new Layer 2 frame and forwards it
- C) Sends the original Layer 2 frame unchanged
- D) Broadcasts the packet to every interface

Q9. Which statement best describes static routing?

- A) Routers automatically exchange routes
- B) Routes are manually configured by the administrator
- C) Routes are learned using hop count
- D) Routes are learned using OSPF

Q10. Which statement best describes dynamic routing?

- A) Routes are manually entered and never change
- B) Routing tables are built and maintained automatically through communication between routers
- C) It works only with wireless networks
- D) It does not use routing protocols

Q11. Which is a routing protocol mentioned in the training?

- A) HTTP
- B) FTP
- C) DNS
- D) OSPF

Q12. A router has two possible routes to the same destination. What does a routing protocol use to help select the best route?

- A) Metric
- B) SSID
- C) MAC vendor
- D) Cable type

Q13. According to the training, the best path is generally selected based on:

- A) Highest cost
- B) Lowest cost
- C) Highest IP address
- D) Oldest route

Q14. Which can be used as a routing metric?

- A) Screen resolution
- B) Username
- C) Hop count
- D) File size

Q15. What does a hop represent in RIP?

- A) A router
- B) A switch port
- C) A network cable
- D) A DNS server

Q16. Which metric does RIP use?

- A) Bandwidth
- B) Delay
- C) Reliability
- D) Hop count

Q17. According to the training, how frequently does RIP send its routing updates?

- A) Every 5 seconds
- B) Every 15 seconds
- C) Every 30 seconds
- D) Every 60 seconds

Q18. What destination address does the training give for RIP's broadcast updates?

- A) 255.255.255.255
- B) 224.0.0.1
- C) 192.168.1.255
- D) 0.0.0.0

Q19. What is equal-cost load balancing?

- A) Blocking all redundant paths
- B) Using multiple paths when they have equal-cost metrics
- C) Using only static routes
- D) Sending every packet through the slowest path

Q20. Why can load balancing improve network performance?

- A) It eliminates all routers
- B) It allows traffic to use multiple equal-cost paths
- C) It removes the need for IP addresses
- D) It converts dynamic routing into static routing

Q21. Scenario: R1 has two routes to Network X: Route A cost 10, Route B cost 20. Which should R1 prefer based on the training's lowest-cost concept?

- A) Route B

- B) Both equally
- C) Route A
- D) Neither route

Q22. Scenario: Network A is directly attached to R1; Network B is reachable only through R2. How are they classified?

- A) A = Remote, B = Directly Connected
- B) A = Directly Connected, B = Remote
- C) Both = Remote
- D) Both = Directly Connected

Q23. Scenario: A company has 200 routers. The administrator manually enters routes for every destination and the network changes frequently. Which approach is generally more suitable?

- A) Static routing
- B) Dynamic routing
- C) No routing
- D) Broadcast routing

Q24. Scenario: R1 has Path A = 2 hops and Path B = 5 hops to the same destination. If RIP is used, which path should be preferred?

- A) Path B
- B) Both paths equally
- C) Path A
- D) The path with the higher IP address

Q25. Which routing protocol is classified as Distance-Vector in the training?

- A) RIP
- B) OSPF
- C) BGP
- D) IS-IS

Q26. Which routing protocol is classified as Link-State in the training?

- A) RIP
- B) OSPF
- C) EIGRP
- D) BGP

Q27. According to the training's classification table, which is listed as Hybrid?

- A) RIP
- B) OSPF
- C) BGP
- D) EIGRP

Q28. What is a key characteristic of Distance-Vector routing in the training?

- A) It uses a map of complete network connectivity
- B) It determines the best route based on the lowest number of hops
- C) It never exchanges routing information
- D) It requires manual configuration of every route

Q29. What is a key characteristic of Link-State routing in the training?

- A) It calculates the best path based on a map of network connectivity
- B) It only counts routers
- C) It uses broadcast every 30 seconds
- D) It requires all routes to be manually entered

Q30. Scenario: A router de-encapsulates the Layer 2 frame, checks the destination IP, selects the best path, then creates a new Layer 2 frame and sends it through the exit interface. Which sequence is correct?

- A) Re-encapsulate → De-encapsulate → Examine IP
- B) Examine IP → Broadcast → De-encapsulate
- C) De-encapsulate → Examine destination IP → Re-encapsulate & Forward
- D) Broadcast → Examine IP → Change destination IP

5. Answer Key

1-B | 2-A | 3-C | 4-B | 5-B

6-C | 7-A | 8-B | 9-B | 10-B

11-D | 12-A | 13-B | 14-C | 15-A

16-D | 17-C | 18-A | 19-B | 20-B

21-C | 22-B | 23-B | 24-C | 25-A

26-B | 27-C | 28-B | 29-A | 30-C

6. Final Memory Card

- Router = connects networks + chooses forwarding path.
- Directly Connected = directly attached to router.
- Remote = reached through another router.
- Static = manual.
- Dynamic = automatic through routing protocols.
- Metric = used to compare routes.
- RIP = hop count; training says updates every 30 seconds.
- Equal-cost paths can be used for load balancing.
- Training classification: RIP/EIGRP = Distance-Vector; OSPF = Link-State; BGP = Hybrid.